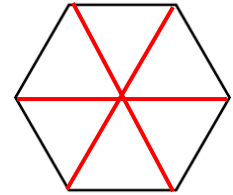


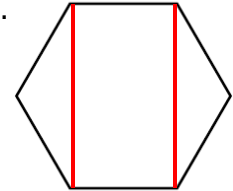
Grade 6 Accelerated
Unit 7
Lesson 7 Homework

Name Answer Key
Date _____
Period _____

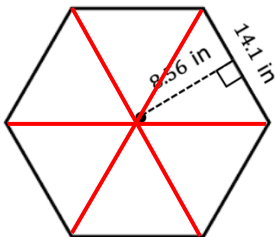
1. Decompose the regular hexagon into only triangles.



2. Decompose the regular hexagon into triangles and quadrilaterals.
Answers may vary.

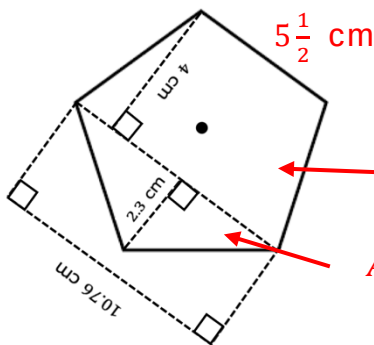


3. A hexagon is given in inches (in). Each side of the hexagon has a length of 14.1 inches and a perpendicular distance from the center to the midpoint of a side to be 8.56 inches. Find the area of the regular hexagon rounded to the nearest square inch.



$$A = 6 \left(\frac{1}{2} \cdot 8.56 \cdot 14.1 \right) = 6(60.348) \\ = 362 \text{ in}^2$$

4. Find the area of the regular pentagon that has a side length of $5\frac{1}{2}$ cm.



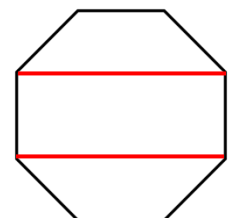
$$A = \frac{1}{2} \cdot 4 \cdot \left(5\frac{1}{2} + 10.76 \right) = 32.52 \text{ cm}^2$$

$$A = \frac{1}{2} \cdot 2.3 \cdot 10.76 = 12.374 \text{ cm}^2$$

$$\text{Total Area} = 32.52 + 12.374 \\ = 44.894 \text{ cm}^2$$

5. Decompose the regular octagon into three quadrilaterals.

Angle of lines may vary.



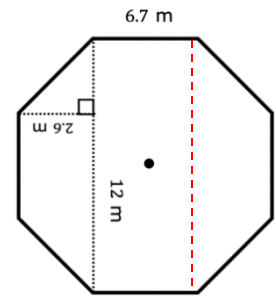
6. An octagon is shown with lengths in meters (m). What is the area of the octagon?

$$A_{trap} = \frac{1}{2} \cdot 2.6 \cdot (6.7 + 12) = 24.31$$

$$2 \text{ trapezoids} \rightarrow 2(24.31) = 48.62$$

$$A_{rect} = 6.7(12) = 80.4$$

$$\text{Total Area} = 48.62 + 80.4 = 129.02 \text{ m}^2$$



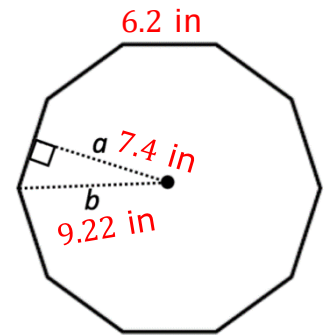
7. Label the regular decagon with the following dimensions.
Side length = 6.2 in.

$$a = 7.4 \text{ in.}$$

$$b = 9.22 \text{ in.}$$

Which of the dimensions is not needed to find the area of the decagon?

b

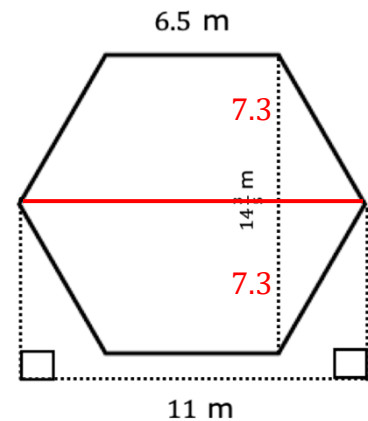


8. Find the area of the decagon from Question 7 in square inches.

$$A = 10 \left(\frac{1}{2} \cdot 7.4 \cdot 6.2 \right) = 229.4 \text{ in}^2$$

9. Myles found the area of the hexagon. His work is shown.

Myles' Work
Area of Trapezoid = $7.3(11 + 6.5)$ $A = 127.75 \text{ m}^2$
Area of hexagon = $2(127.75)$ $A = 255.5 \text{ m}^2$



Write a note to Myles explaining the error he made.

The area of a trapezoid is found by taking half of the height multiplied by the sum of the bases. He forgot to multiply by $\frac{1}{2}$.

10. Find the area of the hexagon in square meters.

$$A_{trap} = \frac{1}{2} \cdot 7.3(11 + 6.5) = 63.875$$

$$\text{Total area} = 2(63.875) = 127.75 \text{ m}^2$$